Posterior Sampling Time Analysis – $m = 64$ and $T = 500$

Simulation Set-Up and Overall Model Fitting Time

When fitting each of our 4 models (`fullGPfixedL`, `NNGPblockFixedL`, `NNGPsequenFixedL`, and `NNGPsequenVaryLj`) with $m = 8^2 = 64$ and $T = 500$, we specified both `equalTimeDist = TRUE` and `equalTimeDist = FALSE`. We thus have $4 \times 2 = 8$ model settings.

Since $m = 64$ is quite small, we do not expect any significant differences in the posterior sampling time between our 4 models with the same `equalTimeDist` input. Since $T = 500$ is quite large, what we have discussed in Appendix B of our manuscript suggest that the 3 temporal parameters' posterior sampling steps, especially the step for $\boldsymbol{\eta}_t$'s, can be markedly accelerated by adopting our approaches and specifying `equalTimeDist = TRUE` for each of our 4 models.

The results we have obtained corroborate these well. With `equalTimeDist = TRUE` specified, the overall model fitting time for our 4 models, i.e., `fullGPfixedL`, `NNGPblockFixedL`, `NNGPsequenFixedL`, and `NNGPsequenVaryLj`, are 7.04 hours, 7.03 hours, 7.04 hours, and 6.31 hours, respectively.[1] If we do not take advantage of our tactics for evenly dispersed time points presented in Appendix B of our manuscript by specifying `equalTimeDist = FALSE` instead, we will need 28.98 days, 29.09 days, 29.03 days, and 28.74 days to fit the same 4 models using the same computation resources.

Portions of Recorded Gibbs Sampler Time for 10 Key Parameters

We first display the first 10 kept post-burn-in MCMC iterations' posterior sampling time (in milliseconds) for 10 key Gibbs sampler steps (corresponding to ψ, Υ, $\boldsymbol{\eta}_t$'s, $z_{jl}(\boldsymbol{s}_{i,o})$'s or $u_j(\boldsymbol{s}_{i,o})$'s, $\xi_j(\boldsymbol{s}_{i,o})$'s, θ_{jl_j}'s, $\delta_{1:k}$, ρ, κ, and $\alpha_{jl_j}(\boldsymbol{s}_{i,o})$'s) for our 8 model settings.

```r
wd <- paste(projDirec, "simu/appen/equalTimeDistAccelerationVerification/m64T500", sep = "/")
setwd(wd)
load("GibbsStepTimeFixedLfullGP.RData")
load("GibbsStepTimeFixedLfullGPfast.RData")
load("GibbsStepTimeFixedLblock.RData")
load("GibbsStepTimeFixedLblockFast.RData")
load("GibbsStepTimeFixedLsequen.RData")
load("GibbsStepTimeFixedLsequenFast.RData")
load("GibbsStepTimeVaryLjSequen.RData")
load("GibbsStepTimeVaryLjSequenFast.RData")
head(GibbsStepTimeFixedLfullGP, 10)      # equalTimeDist = FALSE
```

```
##        z xi theta delta alpha kappa rho   eta upsilon psi
##  [1,] 90 33   296     1     2     1   1 71014       3 128
##  [2,] 85 33   298     1     3     1   1 71013       3 126
##  [3,] 85 33   312     1     3     1   1 71174       3 124
##  [4,] 86 32   297     1     2     1   1 71167       3 182
##  [5,] 84 33   301     1     3     1   1 79008       3 135
##  [6,] 87 31   291     1     3     1   1 79103       3  83
##  [7,] 84 32   305     1     3     1   1 79138       3 180
```

[1] It is worth noting that we utilized a version of our package with the extra step-wise time recording code portions to fit the above models. This package version is slower than our original package version presented in the GitHub repository.

```
##  [8,] 88 32   306     1     3     1   1 79408       4  76
##  [9,] 84 32   301     1     2     1   1 79383       3 175
## [10,] 87 32   305     1     3     1   1 79103       3 135
```

```r
head(GibbsStepTimeFixedLfullGP.fast, 10) # equalTimeDist = TRUE
```

```
##        z xi theta delta alpha kappa rho eta upsilon psi
##  [1,] 82 32   317     1     3     1   2 327       2  53
##  [2,] 83 34   307     1     3     1   1 330       2  54
##  [3,] 81 32   306     1     3     1   1 322       2  53
##  [4,] 82 33   305     1     3     1   1 323       2  54
##  [5,] 83 34   309     1     3     1   1 324       2  53
##  [6,] 81 33   318     1     3     1   1 329       2  53
##  [7,] 83 33   326     1     3     1   1 329       2  54
##  [8,] 85 32   319     1     3     1   1 327       2  56
##  [9,] 82 33   315     1     3     1   1 325       2  54
## [10,] 84 32   311     1     3     1   2 323       2  54
```

```r
head(GibbsStepTimeFixedLblock, 10)       # equalTimeDist = FALSE
```

```
##        z xi theta delta alpha kappa rho   eta upsilon psi
##  [1,] 78 33   304     1     3     1   1 70051       3  89
##  [2,] 82 33   300     1     3     1   1 70058       3 175
##  [3,] 80 32   289     1     3     1   1 72111       3 127
##  [4,] 79 31   295     1     3     1   1 72278       3 183
##  [5,] 78 32   294     1     3     1   1 76961       4 110
##  [6,] 79 32   306     1     3     1   1 77133       4 125
##  [7,] 80 32   300     1     3     1   1 77064       4  89
##  [8,] 77 31   296     1     3     1   1 76953       4 139
##  [9,] 78 32   292     1     3     1   1 76967       4 180
## [10,] 78 32   301     1     3     1   1 75239       4 137
```

```r
head(GibbsStepTimeFixedLblock.fast, 10)  # equalTimeDist = TRUE
```

```
##        z xi theta delta alpha kappa rho eta upsilon psi
##  [1,] 82 35   325     1     3     1   1 324       2  52
##  [2,] 85 34   322     1     3     1   1 324       2  51
##  [3,] 82 33   314     1     3     1   1 313       2  51
##  [4,] 83 35   317     1     3     1   2 315       2  52
##  [5,] 81 32   318     1     3     1   1 323       2  52
##  [6,] 86 34   313     1     3     1   1 330       2  54
##  [7,] 84 35   307     1     3     1   1 327       2  53
##  [8,] 81 33   312     1     3     1   1 329       2  54
##  [9,] 83 34   314     1     3     1   1 325       2  55
## [10,] 84 32   315     1     3     1   1 326       2  54
```

```r
head(GibbsStepTimeFixedLsequen, 10)      # equalTimeDist = FALSE
```

```
##        z xi theta delta alpha kappa rho   eta upsilon psi
##  [1,] 79 33   296     1     9     1   1 75497       4 123
##  [2,] 82 33   295     1     9     1   1 75417       4  82
##  [3,] 81 32   298     1     9     1   1 75276       4 133
##  [4,] 80 32   300     1     9     1   1 75807       4 143
##  [5,] 80 33   301     1     9     1   1 79764       4 142
##  [6,] 79 34   310     1     9     1   1 75835       4  89
##  [7,] 80 33   317     1     9     1   1 75846       3 125
##  [8,] 82 34   308     1     9     1   1 75104       4 180
```

```
##  [9,] 81 33   309     1     9     1   1 79590       3  93
## [10,] 81 31   305     1     9     1   1 79517       3 175
```

```
head(GibbsStepTimeFixedLsequen.fast, 10) # equalTimeDist = TRUE
```

```
##        z xi theta delta alpha kappa rho eta upsilon psi
##  [1,] 75 33   321     1     9     1   1 327       2  57
##  [2,] 77 34   318     1     9     1   1 325       2  57
##  [3,] 80 33   322     1     9     1   1 329       2  56
##  [4,] 73 33   316     1    12     4   3 336       6  60
##  [5,] 77 32   316     1     9     1   1 331       2  56
##  [6,] 77 34   319     1     9     2   1 328       2  53
##  [7,] 76 32   327     1     9     1   1 330       2  54
##  [8,] 76 33   322     1     9     1   1 324       2  53
##  [9,] 79 34   321     1     9     1   1 326       2  53
## [10,] 79 33   328     1     9     1   1 328       2  54
```

```
head(GibbsStepTimeVaryLjSequen, 10)      # equalTimeDist = FALSE
```

```
##       u xi theta delta alpha kappa rho   eta upsilon psi
##  [1,] 0  6   293     1    27     1   1 79983       4 114
##  [2,] 0  6   311     1    27     1   1 80032       3 161
##  [3,] 0  6   295     1    27     0   1 67313       4 112
##  [4,] 1  6   301     1    28     1   1 67432       4 122
##  [5,] 0  6   296     1    27     1   1 70965       3 113
##  [6,] 0  6   299     1    27     1   1 75507       3 106
##  [7,] 0  6   299     1    28     1   1 70782       3 130
##  [8,] 0  6   304     1    27     1   1 72513       4  97
##  [9,] 1  6   307     1    28     1   1 66166       3  93
## [10,] 0  6   293     1    27     1   1 72674       3  90
```

```
head(GibbsStepTimeVaryLjSequen.fast, 10) # equalTimeDist = TRUE
```

```
##       u xi theta delta alpha kappa rho eta upsilon psi
##  [1,] 1  7   310     1    33     1   1 324       2  54
##  [2,] 1  7   320     1    32     1   1 331       2  54
##  [3,] 1  7   314     1    30     1   1 326       2  54
##  [4,] 1  7   317     1    32     1   1 327       2  53
##  [5,] 1  8   319     1    33     1   1 324       2  54
##  [6,] 1  7   317     1    40     1   1 322       2  52
##  [7,] 1  7   313     1    30     1   1 328       2  55
##  [8,] 1  7   319     1    32     1   1 327       2  54
##  [9,] 1  7   312     1    31     1   1 323       2  55
## [10,] 1  7   317     1    31     1   1 329       2  56
```

Posterior Sampling Time Summary Statistics for the 3 Temporal Parameters

We then present vital posterior sampling time summary statistics for the 3 temporal parameters ψ, Υ, and $\boldsymbol{\eta}_t$'s to showcase the manifest computational acceleration brought about by our tactics for evenly dispersed time points presented in Appendix B.

```
fullGPfixedL.slow <- apply(GibbsStepTimeFixedLfullGP[,8:10], 2, summary)
fullGPfixedL.fast <- apply(GibbsStepTimeFixedLfullGP.fast[,8:10], 2, summary)
NNGPblockFixedL.slow <- apply(GibbsStepTimeFixedLblock[,8:10], 2, summary)
```

```r
NNGPblockFixedL.fast <- apply(GibbsStepTimeFixedLblock.fast[,8:10], 2, summary)
NNGPsequenFixedL.slow <- apply(GibbsStepTimeFixedLsequen[,8:10], 2, summary)
NNGPsequenFixedL.fast <- apply(GibbsStepTimeFixedLsequen.fast[,8:10], 2, summary)
NNGPsequenVaryLj.slow <- apply(GibbsStepTimeVaryLjSequen[,8:10], 2, summary)
NNGPsequenVaryLj.fast <- apply(GibbsStepTimeVaryLjSequen.fast[,8:10], 2, summary)
fullGPfixedLsummary <- data.frame(eta = fullGPfixedL.slow[,1],
                                  eta.fast = fullGPfixedL.fast[,1],
                                  upsilon = fullGPfixedL.slow[,2],
                                  upsilon.fast = fullGPfixedL.fast[,2],
                                  psi = fullGPfixedL.slow[,3],
                                  psi.fast = fullGPfixedL.fast[,3])
NNGPblockFixedLsummary <- data.frame(eta = NNGPblockFixedL.slow[,1],
                                     eta.fast = NNGPblockFixedL.fast[,1],
                                     upsilon = NNGPblockFixedL.slow[,2],
                                     upsilon.fast = NNGPblockFixedL.fast[,2],
                                     psi = NNGPblockFixedL.slow[,3],
                                     psi.fast = NNGPblockFixedL.fast[,3])
NNGPsequenFixedLsummary <- data.frame(eta = NNGPsequenFixedL.slow[,1],
                                      eta.fast = NNGPsequenFixedL.fast[,1],
                                      upsilon = NNGPsequenFixedL.slow[,2],
                                      upsilon.fast = NNGPsequenFixedL.fast[,2],
                                      psi = NNGPsequenFixedL.slow[,3],
                                      psi.fast = NNGPsequenFixedL.fast[,3])
NNGPsequenVaryLjsummary <- data.frame(eta = NNGPsequenVaryLj.slow[,1],
                                      eta.fast = NNGPsequenVaryLj.fast[,1],
                                      upsilon = NNGPsequenVaryLj.slow[,2],
                                      upsilon.fast = NNGPsequenVaryLj.fast[,2],
                                      psi = NNGPsequenVaryLj.slow[,3],
                                      psi.fast = NNGPsequenVaryLj.fast[,3])
fullGPfixedLsummary
```

```
##              eta eta.fast upsilon upsilon.fast      psi psi.fast
## Min.    47910.00 307.0000  3.0000       2.0000  68.0000  51.0000
## 1st Qu. 71600.25 326.0000  3.0000       2.0000 114.0000  54.0000
## Median  74556.00 333.0000  3.0000       2.0000 129.0000  55.0000
## Mean    73937.04 334.2716  3.4484       2.0234 131.1978  54.9572
## 3rd Qu. 77246.50 342.0000  4.0000       2.0000 152.2500  56.0000
## Max.    81935.00 403.0000  5.0000       6.0000 192.0000  73.0000
```

```r
NNGPblockFixedLsummary
```

```
##              eta eta.fast upsilon upsilon.fast      psi psi.fast
## Min.    49929.00 309.0000  3.0000       2.0000  69.0000  51.0000
## 1st Qu. 72014.75 326.0000  3.0000       2.0000 114.0000  54.0000
## Median  75084.00 333.0000  3.0000       2.0000 129.0000  55.0000
## Mean    74305.95 334.8152  3.4336       2.0162 131.7338  55.0994
## 3rd Qu. 77423.25 342.0000  4.0000       2.0000 157.0000  56.0000
## Max.    81945.00 417.0000  5.0000       5.0000 193.0000  72.0000
```

```r
NNGPsequenFixedLsummary
```

```
##               eta eta.fast upsilon upsilon.fast      psi psi.fast
## Min.     51226.00 307.0000  3.0000       2.0000  69.0000  50.0000
## 1st Qu.  72183.25 326.0000  3.0000       2.0000 116.0000  54.0000
## Median   75226.00 333.0000  4.0000       2.0000 131.0000  55.0000
```

```
## Mean     76910.86 335.0714  3.5734       2.0292 136.7524  55.1294
## 3rd Qu.  78106.50 342.0000  4.0000       2.0000 164.0000  56.0000
## Max.    129933.00 415.0000  6.0000       6.0000 285.0000  72.0000
```

```
NNGPsequenVaryLjsummary
```

```
##               eta eta.fast upsilon upsilon.fast      psi psi.fast
## Min.     42653.00 306.0000  3.0000       2.0000  68.0000   51.000
## 1st Qu.  70463.75 323.0000  3.0000       2.0000 113.0000   54.000
## Median   74632.50 328.0000  4.0000       2.0000 129.0000   54.000
## Mean     75795.19 330.1242  3.5962       2.0008 133.8518   54.725
## 3rd Qu.  77780.25 335.0000  4.0000       2.0000 159.0000   56.000
## Max.    134470.00 390.0000  6.0000       4.0000 295.0000   69.000
```

The results correspond well to what we have discussed in Appendix B of our manuscript. **For each of our 4 models, specifying `equalTimeDist = TRUE` instead of `equalTimeDist = FALSE` on equispaced time points markedly accelerates posterior Gibbs sampler steps of η_t's, ψ, and Υ (especially the step for η_t's).**

We finally calculate standard deviations for the 3 temporal-related parameters' posterior sampling time across all kept post-burn-in MCMC iterations.

```
round(apply(GibbsStepTimeFixedLfullGP[,8:10], 2, sd), 5)
```

```
##        eta    upsilon        psi
## 4305.40218    0.49859   28.33158
```

```
round(apply(GibbsStepTimeFixedLfullGP.fast[,8:10], 2, sd), 5)
```

```
##      eta  upsilon      psi
## 11.53834  0.22375  1.98357
```

```
round(apply(GibbsStepTimeFixedLblock[,8:10], 2, sd), 5)
```

```
##        eta    upsilon        psi
## 4412.38304    0.49602   28.56903
```

```
round(apply(GibbsStepTimeFixedLblock.fast[,8:10], 2, sd), 5)
```

```
##      eta  upsilon      psi
## 11.68779  0.15473  1.96598
```

```
round(apply(GibbsStepTimeFixedLsequen[,8:10], 2, sd), 5)
```

```
##         eta     upsilon         psi
## 10030.28478     0.62824    33.26132
```

```
round(apply(GibbsStepTimeFixedLsequen.fast[,8:10], 2, sd), 5)
```

```
##      eta  upsilon      psi
## 12.17556  0.25992  2.00416
```

```
round(apply(GibbsStepTimeVaryLjSequen[,8:10], 2, sd), 5)
```

```
##         eta     upsilon         psi
## 10332.05841     0.63626    34.01907
```

```
round(apply(GibbsStepTimeVaryLjSequen.fast[,8:10], 2, sd), 5)
```

```
##      eta  upsilon      psi
## 10.41849  0.03464  1.77392
```